Lecturer: **Approved by:**

. . . . . .

UNIVERSITY OF TECHNOLOGY VNUHCM FACULTY OF AS	FINAL EXAM	Semester/ Academic year		202	2020 - 2021
		Date	08 May 2021		
	Course title	Calculus 1			
	Course ID	MT1003			
	Duration	90 mins	Question sheet code	1254	

Intructions to students: - There are 5 pages in the exam
-This is a closed book exam. Only your calculator is allowed. Total available score: 10.
-At the beginning of the working time, you MUST fill in your full name and student ID on this question sheet.
-All essential steps of calculations, analyses, justifications and final results are required for full credit.
Any answer without essential calculation steps, and/or analyses, and/or justifications will earn zero mark.

Student's full name: . . . . . .

Student ID: . . . . . .

Invigilator 1:. . . . . .

Invigilator 2:. . . . . .

Part I. Multiple choice (6 points, 60 minutes)

Question 01. If $f(x) = \int_0^x \cos(t^2)dt$, find $A = \frac{d}{dx}f(\sqrt{x})$.

(A) $A = \frac{\cos x}{3\sqrt{x}}$ (B) $A = \frac{\sin x}{3\sqrt{x}}$ (C) $A = \frac{\cos x}{2\sqrt{x}}$ (D) $A = \frac{\sin x}{2\sqrt{x}}$

Question 02. Find the approximation of c such that

$$\int_c^x f(t)dt = x^3 + 3x^2 + 2x - 3.$$

(A) $c = 0.9717$ (B) $c = 0.8717$ (C) $c = 0.6717$ (D) $c = 0.7717$

Question 03. If $F(x) = x^2$ is an antiderivative of the function $f(x)e^{2x}$. Find the antiderivative of the function $f'(x)e^{2x}$.

(A) None of them (B) $\int f'(x)e^{2x}\mathrm{d}x = -x^2 + x + C$

(C) $\int f'(x)e^{2x}\mathrm{d}x = x^2 - 2x + C$ (D) $\int f'(x)e^{2x}\mathrm{d}x = -2x^2 + 2x + C$

Question 04. If $F(x)$ is an antiderivative of the function $f(x) = \frac{\ln x}{x}\cdot$ Calculate $I = F(\mathrm{e}) - F(1)$.

(A) $I = \frac{1}{\mathrm{e}}$ (B) $I = \mathrm{e}$ (C) None of them (D) $I = \frac{1}{2}$

Question 05. If the functions f and g have continuous derivative on $[0, 2]$ and satisfies the condition $\int_0^2 f'(x)g(x)\,\mathrm{d}x = 2$, $\int_0^2 f(x)g'(x)\,\mathrm{d}x = 3$, then calculate $\int_0^2 \left[f(x)g(x)\right]'\mathrm{d}x$.

(A) $I = 5$ (B) $I = 0$ (C) $I = 6$ (D) $I = 1$

Question 06. If $\int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x = 5$, then calculate $I = \int_0^{\frac{\pi}{2}} [f(x) + 2\sin x]\,\mathrm{d}x$.

(A) $5 + \frac{\pi}{2}$ (B) 3 (C) 7 (D) $5 + \pi$

Question 07. If the function $f(x)$ has continuous derivative on $[0,1]$ and satisfies the condition $f(0) = f(1) = 1$, and $\int_0^1 e^x \left[f(x) + f'(x)\right] \mathrm{d}x = ae + b$. Calculate $Q = a^{2018} + b^{2018}$.

(A) $Q = 2^{2017} - 1$ (B) $Q = 2^{2017} + 1$ (C) $Q = 0$ (D) $Q = 2$

Question 08. When a particle is located a distance x meter from the origin, a force of given $F(x) = \frac{1}{\sqrt{9 + 8x - x^2}}$ (newton) acts on it. How much work is done in moving it from $x = \frac{13}{2}$ to $x = 9$.

(A) $W = \frac{4\pi}{3}$ (B) $W = \frac{\pi}{3}$ (C) $W = \frac{2\pi}{3}$ (D) $W = \frac{\pi}{4}$

Question 09. When a particle is located a distance x meter from the origin, a force of given $F(x) = \frac{2x^2 + 3x + 7}{x^3 + x^2 - x - 1}$ (newton) acts on it. Find the approximation of the work done in moving it from $x = 4$ to $x = 10$, rounded to two decimal numbers.

(A) None of them (B) $W \approx 3.18$ (C) $W \approx 4.18$ (D) $W \approx 2.18$

Question 10. When a particle is located a distance x meter from the origin, a force of given $F(x) = \frac{2x^2 - 5x + 2}{x^3 + x}$ (newton) acts on it. How much work is done in moving it from $x = 1$ to $x = 5$.

(A) $W = 3\ln 5 - 5\arctan 5 + \frac{5\pi}{4}$ (B) $W = 3\ln 5 - 5\arctan 5 + \frac{9\pi}{4}$

(C) $W = 2\ln 5 - 5\arctan 5 + \frac{5\pi}{4}$ (D) $W = 2\ln 5 - 5\arctan 5 + \frac{7\pi}{4}$

Question 11. Evaluate displacement of one particle with velocity $v(t) = \frac{1}{t^2} \cdot e^{-\frac{1}{t}}$ (m/s) from $t = 1s$ to $t = 2s$.

(A) $D = 2e^{-\frac{1}{2}} - 3e^{-1}$ (B) $D = e^{-\frac{1}{2}} - 2e^{-1}$ (C) $D = e^{-\frac{1}{2}} - e^{-1}$ (D) $D = 2e^{-\frac{1}{2}} - e^{-1}$

Question 12. Evaluate displacement of one particle with velocity $v(t) = t + t\cos t \quad (m/s)$ from $t = 0(s)$ to $t = \pi/2(s)$.

(A) None of them (B) $D = \frac{3\pi^2}{8} + \frac{\pi}{2} - 1$ (C) $D = \frac{\pi^2}{8} + \frac{\pi}{2} - 1$ (D) $D = \frac{\pi^2}{8} + \frac{3\pi}{2} - 1$

Question 13. Find the approximation of the surface area generated when the graph of the function $y = \frac{1}{4}x^4 + \frac{1}{8}x^{-2}$ for $x \in [1,2]$ is revolved about the $x-$axis.

(A) $A = 53.88$ (B) $A = 52.88$ (C) $A = 51.88$ (D) $A = 54.88$

Question 14. Find the area of the region bounded by the $x-$axis and the graph of the function $f(x) = \frac{e^x}{(1 + e^x)^2}$

(A) $A = 2$ (B) $A = 1$ (C) $A = 4$ (D) $A = 3$

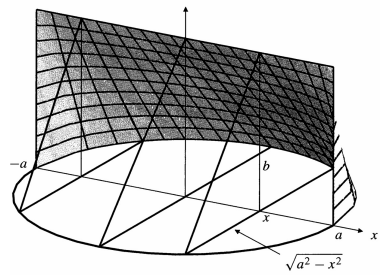
Question 15. Find the volume of the solid obtained by rotating the region bounded by the curve $y = -x^2 + 6x - 8, y = 0$ about the $y-$axis.

(A) None of them (B) $V = 8\pi$ (C) $V = 6\pi$ (D) $V = 10\pi$

Question 16. Find volume of solid with its cross section area $A(x) = \frac{1}{x(x+1)^2}; 1 \leqslant x \leqslant 2$.

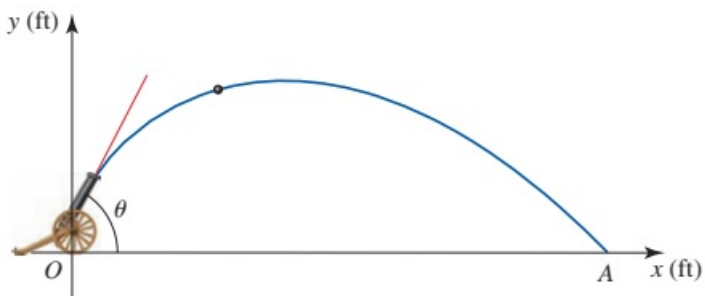
(A) $V = \ln\frac{4}{3} - \frac{5}{6}$ (B) None of them (C) $V = 2\ln\frac{4}{3} - \frac{1}{6}$ (D) $V = \ln\frac{4}{3} - \frac{1}{6}$

Question 17. A tent has a circular base of radius $a = 2$ metres and is supported by a horizontal ridge bar held at height $b = 3$ metres above a diameter of the base by vertical supports at each end of the diameter. The material of the tent is stretched tight so that each cross-section perpendicular to the ridge bar is an isosceles triangle. Find the volume of the tent.



(A) $V = \pi$ (B) $V = 3\pi$ (C) None of them (D) $V = 6\pi$

Question 18. A projectile is fired from a cannon located on a horizontal plane. If we think of the cannon as being located at the origin O of an $xy-$coordinate system, then the path of the projectile is $y = \sqrt{3}.x - \frac{x^2}{400}$; where x and y are measured in feet. Estimate the distance traveled by the projectile in the air, rounded to two decimal numbers.



(A) $L \approx 955.21$ (B) $L \approx 957.21$ (C) None of them (D) $L \approx 956.21$

Question 19. Calculate $I = \int_0^{+\infty} \frac{\sqrt{1+x}}{2+7x}\,\mathrm{d}x$.

(A) $I = 0$ (B) $I = +\infty$ (C) $I = \ln 2$ (D) $I = \ln 3$

Question 20. Calculate $I = \int_0^4 \frac{dx}{(8-x)\sqrt{4-x}}$.

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $+\infty$

Part II. Essay (4 points, 30 minutes)

Question 01. The Pacific halibut fishery has been modeled by the differential equation

$$\frac{dy}{dt} = ky\left(1 - \frac{y}{M}\right),$$

where $y(t)$ is the biomass (the total mass of the members of the population) in kilograms at time t (measured in years), the carrying capacity is estimated to be $M = 8 \times 10^7$ kg, and $k = 0.71$ per year.

(a) If $y(0) = 2 \times 10^7$ kg, find the biomass a year later.

(b) How long will it take for the biomass to reach 4×10^7 kg?

Question 02. Solve the differential equation $y'' - 5y' + 6y = 6e^{4x}$

SOLUTION.

Lecturer: **Approved by:**

. . . . . .

UNIVERSITY OF TECHNOLOGY VNUHCM FACULTY OF AS	FINAL EXAM	Semester/ Academic year		202	2020 - 2021
		Date	08 May 2021		
	Course title	Calculus 1			
	Course ID	MT1003			
	Duration	90 mins	Question sheet code	1254	

Intructions to students: - There are 5 pages in the exam
-This is a closed book exam. Only your calculator is allowed. Total available score: 10.
-At the beginning of the working time, you MUST fill in your full name and student ID on this question sheet.
-All essential steps of calculations, analyses, justifications and final results are required for full credit.
Any answer without essential calculation steps, and/or analyses, and/or justifications will earn zero mark.

Student's full name: . . . . . .

Student ID: . . . . . .

Invigilator 1:. . . . . .

Invigilator 2:. . . . . .

ANSWER

Part I. Multiple choice (6 points, 60 minutes)

01. C	**03.** D	**05.** A	**07.** D	**09.** D	**11.** C	**13.** C	**15.** B	**17.** D	**19.** B
02. C	**04.** D	**06.** C	**08.** B	**10.** C	**12.** C	**14.** B	**16.** D	**18.** D	**20.** A

Part II. Essay (4 points, 30 minutes)

Question 01. Solution. Using separable method, we obtain

$$\frac{dy}{dt} = ky\left(1 - \frac{y}{M}\right) \Rightarrow \frac{dy}{y\left(1 - \frac{y}{M}\right)} = kdt \Rightarrow \int \frac{dy}{y\left(1 - \frac{y}{M}\right)} = \int kdt$$

$$\Rightarrow \int \frac{dy}{y} + \frac{1}{M}\int \frac{dy}{1 - \frac{y}{M}} = k\int dt \Rightarrow \ln|y| - \ln\left|1 - \frac{y}{M}\right| = kt + \ln C$$

$$\Rightarrow \frac{y}{\left|1 - \frac{y}{M}\right|} = Ce^{kt}$$

If $t = 0$, then $\dfrac{y(0)}{\left|1 - \frac{y(0)}{M}\right|} = Ce^{0k} = C \Rightarrow C = \dfrac{8}{3} \times 10^7$.

If $t = 1$, then $\dfrac{y(1)}{\left|1 - \frac{y(1)}{M}\right|} = Ce^{k} \Rightarrow y(1) = \dfrac{Ce^k}{1 + \frac{Ce^k}{M}} \approx 32324112.68$

If $y = 4 \times 10^7$, then $\dfrac{4 \times 10^7}{1 - \dfrac{4 \times 10^7}{M}} = \dfrac{8}{3} \times 10^7 e^{kt}$

$$\Rightarrow e^{kt} = \frac{3 \times 4 \times 10^7}{8 \times 10^7 \times \left(1 - \dfrac{4 \times 10^7}{M}\right)} = 3 \Rightarrow t = \frac{\ln(3)}{k} \approx 1.5473(\text{year})$$

Question 02. Solution.

(a) **Step 1.** Solve the homogeneous equation

$$y'' - 5y' + 6y = 0$$

The characteristic equation $k^2 - 5k + 6 = 0$ has 2 real different roots

$$k_1 = 2, k_2 = 3$$

(b) **Step 2.** The homogeneous solution is

$$y_h = C_1 e^{2x} + C_2 e^{3x}$$

(c) **Step 3.** Find a particular solution of nonhomogeneous equation

$$y'' - 5y' + 6y = 6e^{4x}.$$

The particular solution has a form $y_p = x^s.e^{4x}.A$. Since $\alpha = 4$ is not the root of characteristic equation then $s = 0$ and $y_p = A.e^{4x}$.

$$\begin{array}{c|ccc}
6\times & y_p & = & Ae^{4x} \\
-5\times & y_p' & = & 4Ae^{4x} \\
1\times & y_p'' & = & 16Ae^{4x} \\
\hline
y_p'' - 5y_p' + 6y_p & = & 2Ae^{4x} = & 6e^{4x}
\end{array}$$

$$\Rightarrow A = 3$$

(d) **Step 4.** The general solution is

$$y_{gen} = y_h + y_p = C_1 e^{2x} + C_2 e^{3x} + 3e^{4x}.$$

Lecturer: **Approved by:**

. . . . . .

UNIVERSITY OF TECHNOLOGY VNUHCM FACULTY OF AS	FINAL EXAM	Semester/ Academic year		202	2020 - 2021
		Date	08 May 2021		
	Course title	Calculus 1			
	Course ID	MT1003			
	Duration	90 mins	Question sheet code	1254	

Intructions to students: - There are 5 pages in the exam
-This is a closed book exam. Only your calculator is allowed. Total available score: 10.
-At the beginning of the working time, you MUST fill in your full name and student ID on this question sheet.
-All essential steps of calculations, analyses, justifications and final results are required for full credit.
Any answer without essential calculation steps, and/or analyses, and/or justifications will earn zero mark.

Student's full name: . . . . . .

Student ID: . . . . . .

Invigilator 1:. . . . . .

Invigilator 2:. . . . . .

FINAL EXAM AND THE ANSWER

Part I. Multiple choice (6 points, 60 minutes)

Question 01. If $f(x) = \int_0^x \cos(t^2)dt$, find $A = \frac{d}{dx}f(\sqrt{x})$.

(A) $A = \frac{\cos x}{3\sqrt{x}}$ (B) $A = \frac{\sin x}{3\sqrt{x}}$ (C) $A = \frac{\cos x}{2\sqrt{x}}$ (D) $A = \frac{\sin x}{2\sqrt{x}}$

Solution. The correct answer (C). Let $u = \sqrt{x}$. Then we have

$$\frac{d}{dx}f(\sqrt{x}) = f'(u).u'(x) = (\cos u^2) \cdot \frac{1}{2\sqrt{x}} = \frac{\cos x}{2\sqrt{x}}$$

□

Question 02. Find the approximation of c such that

$$\int_c^x f(t)dt = x^3 + 3x^2 + 2x - 3.$$

(A) $c = 0.9717$ (B) $c = 0.8717$ (C) $c = 0.6717$ (D) $c = 0.7717$

Solution. The correct answer (C). Differentiating both sides of the equation, we obtain

$$f(x) = 3x^2 + 6x + 2 \Rightarrow \int_c^x f(t)dt = \left[t^3 + 3t^2 + 2t\right]_c^x =$$
$$= (x^3 + 3x^2 + 2x) - (c^3 + 3c^2 + 2c) = x^3 + 3x^2 + 2x - 3$$
$$\Rightarrow c^3 + 3c^2 + 2c = 3 \Rightarrow c \approx 0.6717$$

□

Question 03. If $F(x) = x^2$ is an antiderivative of the function $f(x)e^{2x}$. Find the antiderivative of the function $f'(x)e^{2x}$.

(A) None of them

(B) $\int f'(x)e^{2x}\mathrm{d}x = -x^2 + x + C$

(C) $\int f'(x)e^{2x}\mathrm{d}x = x^2 - 2x + C$

(D) $\int f'(x)e^{2x}\mathrm{d}x = -2x^2 + 2x + C$

Solution. The correct answer (D). $F(x) = x^2$ is an antiderivative of the function $f(x)e^{2x}$ so $2x = f(x)e^{2x}$ Let $\begin{cases} u = e^{2x} \\ \mathrm{d}v = f'(x)\mathrm{d}x \end{cases} \Rightarrow \begin{cases} \mathrm{d}u = 2e^{2x}\mathrm{d}x \\ v = f(x) \end{cases}$

$\Rightarrow \int f'(x)e^{2x}\mathrm{d}x = f(x)e^{2x} - 2\int f(x)e^{2x}\mathrm{d}x = 2x - 2x^2 + C$ □

Question 04. If $F(x)$ is an antiderivative of the function $f(x) = \dfrac{\ln x}{x}\cdot$ Calculate $I = F(\mathrm{e}) - F(1)$.

(A) $I = \dfrac{1}{\mathrm{e}}$

(B) $I = \mathrm{e}$

(C) None of them

(D) $I = \dfrac{1}{2}$

Solution. The correct answer (D). We have $I = \displaystyle\int_1^{\mathrm{e}} \frac{\ln x}{x}\,\mathrm{d}x = \int_1^{\mathrm{e}} \ln x\,\mathrm{d}(\ln x) = \left.\frac{(\ln x)^2}{2}\right|_1^{\mathrm{e}} = \frac{1}{2}$. □

Question 05. If the functions f and g have continuous derivative on $[0,2]$ and satisfies the condition $\displaystyle\int_0^2 f'(x)g(x)\,\mathrm{d}x = 2$, $\displaystyle\int_0^2 f(x)g'(x)\,\mathrm{d}x = 3$, then calculate $\displaystyle\int_0^2 \left[f(x)g(x)\right]'\mathrm{d}x$.

(A) $I = 5$

(B) $I = 0$

(C) $I = 6$

(D) $I = 1$

Solution. The correct answer (A). We have $\displaystyle\int_0^2 \left[f(x)g(x)\right]'\mathrm{d}x = \int_0^2 f'(x)g(x)\,\mathrm{d}x + \int_0^2 f(x)g'(x)\,\mathrm{d}x = 2 + 3 = 5$. □

Question 06. If $\displaystyle\int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x = 5$, then calculate $I = \displaystyle\int_0^{\frac{\pi}{2}} [f(x) + 2\sin x]\,\mathrm{d}x$.

(A) $5 + \dfrac{\pi}{2}$

(B) 3

(C) 7

(D) $5 + \pi$

Solution. The correct answer (C). $I = \displaystyle\int_0^{\frac{\pi}{2}} [f(x) + 2\sin x]\,\mathrm{d}x = \int_0^{\frac{\pi}{2}} f(x)\,\mathrm{d}x + \int_0^{\frac{\pi}{2}} 2\sin x\,\mathrm{d}x = 5 - 2\cos x\Big|_0^{\frac{\pi}{2}} = 7$ □

Question 07. If the function $f(x)$ has continuous derivative on $[0,1]$ and satisfies the condition $f(0) = f(1) = 1$, and $\displaystyle\int_0^1 e^x\left[f(x) + f'(x)\right]\mathrm{d}x = ae + b$. Calculate $Q = a^{2018} + b^{2018}$.

(A) $Q = 2^{2017} - 1$

(B) $Q = 2^{2017} + 1$

(C) $Q = 0$

(D) $Q = 2$

Solution. The correct answer (D). We have $\displaystyle\int_0^1 e^x\left[f(x) + f'(x)\right]\mathrm{d}x = \int_0^1 \left[e^x f(x)\right]'\mathrm{d}x = e^x f(x)\Big|_0^1 = ef(1) - f(0) = e - 1$. Thus $\begin{cases} a = 1 \\ b = -1 \end{cases}$ Therefore, $Q = 2$. □

Question 08. When a particle is located a distance x meter from the origin, a force of given $F(x) = \dfrac{1}{\sqrt{9 + 8x - x^2}}$ (newton) acts on it. How much work is done in moving it from $x = \dfrac{13}{2}$ to $x = 9$.

(A) $W = \dfrac{4\pi}{3}$

(B) $W = \dfrac{\pi}{3}$

(C) $W = \dfrac{2\pi}{3}$

(D) $W = \dfrac{\pi}{4}$

Solution. The correct answer (B). $W = \int_{13/2}^{9} F(x)dx = \int_{13/2}^{9} \frac{dx}{\sqrt{9+8x-x^2}} = \int_{13/2}^{9} \frac{d(x-4)}{\sqrt{25-(x-4)^2}} = \arcsin \frac{(x-4)}{5}\Big|_{13/2}^{9} =$ $\arcsin 1 - \arcsin \frac{1}{2} = \frac{\pi}{2} - \frac{\pi}{6} = \frac{\pi}{3}$. □

Question 09. When a particle is located a distance x meter from the origin, a force of given $F(x) = \frac{2x^2+3x+7}{x^3+x^2-x-1}$ (newton) acts on it. Find the approximation of the work done in moving it from $x = 4$ to $x = 10$, rounded to two decimal numbers.

(A) None of them (B) $W \approx 3.18$ (C) $W \approx 4.18$ (D) $W \approx 2.18$

Solution. The correct answer (D). Partial decomposition gives us

$$F(x) = \frac{2x^2+3x+7}{x^3+x^2-x-1} = \frac{2x^2+3x+7}{(x+1)^2(x-1)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x-1}$$

$$\Rightarrow 2x^2+3x+7 \equiv A(x-1)(x+1) + B(x-1) + C(x+1)^2, \forall x \in \mathbb{R}.$$

When $x = 1 \Rightarrow 2 \times 1^2 + 3 \times 1 + 7 = C(1+1)^2 \Rightarrow C = 3$.
When $x = -1 \Rightarrow 2 \times (-1)^2 + 3 \times (-1) + 7 = B(-1-1) \Rightarrow B = -3$.
When $x = 0 \Rightarrow 2 \times 0^2 + 3 \times 0 + 7 = A(0-1)(0+1) + B(0-1) + C(0+1)^2 \Rightarrow A = 1$.
The work done in moving it from $x = 4$ to $x = 10$ is

$$W = \int_4^{10} F(x)dx = \int_4^{10} \left(\frac{1}{x+1} - \frac{3}{(x+1)^2} + \frac{3}{x-1} \right) dx =$$

$$= \left[\ln|x+1| + \frac{3}{x+1} + 3\ln|x-1| \right]_4^{10} = 3\ln 3 - \ln 11 + \ln 5 - \frac{18}{55} \approx 2.18.$$

□

Question 10. When a particle is located a distance x meter from the origin, a force of given $F(x) = \frac{2x^2-5x+2}{x^3+x}$ (newton) acts on it. How much work is done in moving it from $x = 1$ to $x = 5$.

(A) $W = 3\ln 5 - 5\arctan 5 + \frac{5\pi}{4}$ (B) $W = 3\ln 5 - 5\arctan 5 + \frac{9\pi}{4}$
(C) $W = 2\ln 5 - 5\arctan 5 + \frac{5\pi}{4}$ (D) $W = 2\ln 5 - 5\arctan 5 + \frac{7\pi}{4}$

Solution. The correct answer (C). The work done is

$$W = \int_1^5 \frac{2x^2-5x+2}{x^3+x} dx = \int_1^5 \left(\frac{2}{x} - \frac{5}{x^2+1} \right) dx = 2\ln 5 - 5\arctan 5 + \frac{5\pi}{4}$$

□

Question 11. Evaluate displacement of one particle with velocity $v(t) = \frac{1}{t^2} \cdot e^{-\frac{1}{t}}$ (m/s) from $t = 1s$ to $t = 2s$.

(A) $D = 2e^{-\frac{1}{2}} - 3e^{-1}$ (B) $D = e^{-\frac{1}{2}} - 2e^{-1}$ (C) $D = e^{-\frac{1}{2}} - e^{-1}$ (D) $D = 2e^{-\frac{1}{2}} - e^{-1}$

Solution. The correct answer (C). The displacement of the particle is

$$D = \int_1^2 v(t)dt = \int_1^2 \frac{1}{t^2} \cdot e^{-\frac{1}{t}} dt = \int_1^2 e^{-\frac{1}{t}} d\left(-\frac{1}{t}\right) = e^{-\frac{1}{t}}\Big|_1^2 = e^{-\frac{1}{2}} - e^{-1}.$$

□

Question 12. Evaluate displacement of one particle with velocity $v(t) = t + t\cos t \quad (m/s)$ from $t = 0(s)$ to $t = \pi/2(s)$.

(A) None of them (B) $D = \frac{3\pi^2}{8} + \frac{\pi}{2} - 1$ (C) $D = \frac{\pi^2}{8} + \frac{\pi}{2} - 1$ (D) $D = \frac{\pi^2}{8} + \frac{3\pi}{2} - 1$

Solution. The correct answer (C). The displacement of the particle is

$$D = \int_0^{\pi/2} v(t)dt = \int_0^{\pi/2} (t + t\cos t)dt = \int_0^{\pi/2} tdt + \int_0^{\pi/2} t\cos tdt.$$

Let $\begin{cases} u = t \\ dv = \cos tdt \end{cases} \Rightarrow \begin{cases} du = dt \\ v = \sin t \end{cases}$

Using integration by parts, we have

$$D = \left[\frac{t^2}{2} + t\sin t + \cos t\right]_0^{\pi/2} = \frac{\pi^2}{8} + \frac{\pi}{2} - 1$$

□

Question 13. Find the approximation of the surface area generated when the graph of the function $y = \frac{1}{4}x^4 + \frac{1}{8}x^{-2}$ for $x \in [1, 2]$ is revolved about the $x-$axis.

(A) $A = 53.88$ (B) $A = 52.88$ (C) $A = 51.88$ (D) $A = 54.88$

Solution. The correct answer (C). The required surface area is

$$A = 2\pi\int_1^2 \left(\frac{1}{4}x^4 + \frac{1}{8}x^{-2}\right).\sqrt{1 + \left(x^3 - \frac{1}{4}x^{-3}\right)^2}dx =$$

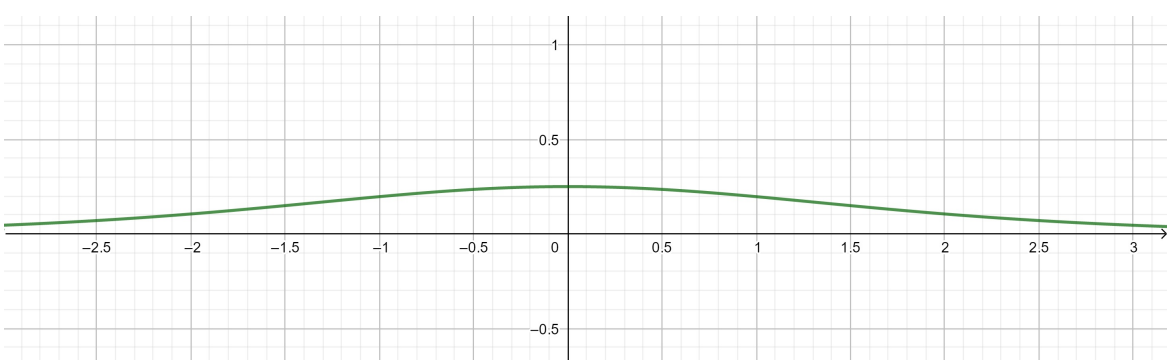
$$= 2\pi\int_1^2 \left(\frac{1}{4}x^4 + \frac{1}{8}x^{-2}\right).\left(x^3 + \frac{1}{4}x^{-3}\right)dx \approx 51.88$$

□

Question 14. Find the area of the region bounded by the $x-$axis and the graph of the function $f(x) = \frac{e^x}{(1+e^x)^2}$

(A) $A = 2$ (B) $A = 1$ (C) $A = 4$ (D) $A = 3$

Solution. The correct answer (B). The graph of function f



The required area is

$$A = \int_{-\infty}^{+\infty} |f(x)|dx = \int_{-\infty}^{+\infty} \frac{e^x dx}{(1+e^x)^2} = \int_{-\infty}^{+\infty} \frac{d(1+e^x)}{(1+e^x)^2} = \left[-\frac{1}{1+e^x}\right]_{-\infty}^{+\infty} = 1.$$

□

Question 15. Find the volume of the solid obtained by rotating the region bounded by the curve $y = -x^2 + 6x - 8, y = 0$ about the $y-$axis.

(A) None of them (B) $V = 8\pi$ (C) $V = 6\pi$ (D) $V = 10\pi$

Solution. The correct answer (B). Points of intersection of the curve $y = -x^2 + 6x - 8$ and the $x-$axis can be found by solving the equation

$$-x^2 + 6x - 8 = 0 \Leftrightarrow \left[\begin{array}{l} x = 2 \\ x = 4 \end{array} \right.$$

The volume of the solid obtained by rotating the region bounded by the curve $y = -x^2 + 6x - 8, y = 0$ about the $y-$axis is

$$V = 2\pi \int_2^4 x(-x^2 + 6x - 8)dx = 2\pi \int_2^4 (-x^3 + 6x^2 - 8x)dx =$$

$$= 2\pi \left[-\frac{x^4}{4} + 6\frac{x^3}{3} - 8\frac{x^2}{2} \right]_2^4 = 8\pi.$$

□

Question 16. Find volume of solid with its cross section area $A(x) = \dfrac{1}{x(x+1)^2}; 1 \leqslant x \leqslant 2.$

(A) $V = \ln\frac{4}{3} - \frac{5}{6}$ (B) None of them (C) $V = 2\ln\frac{4}{3} - \frac{1}{6}$ (D) $V = \ln\frac{4}{3} - \frac{1}{6}$

Solution. The correct answer (D). We have

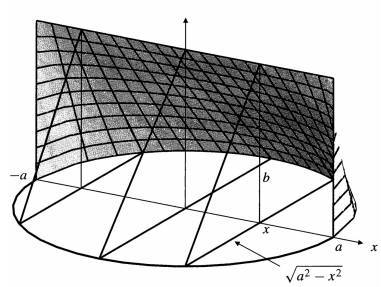
$$V = \int_1^2 A(x)dx = \int_1^2 \frac{1}{x(x+1)^2}dx = \int_1^2 \frac{(x+1) - x}{x(x+1)^2}dx =$$

$$= \int_1^2 \left[\frac{1}{x(x+1)} - \frac{1}{(x+1)^2} \right] dx = \int_1^2 \left[\frac{1}{x} - \frac{1}{x+1} - \frac{1}{(x+1)^2} \right] dx =$$

$$= \left[\ln|x| - \ln|x+1| + \frac{1}{x+1} \right]_1^2 = \ln 2 - \ln\frac{3}{2} - \frac{1}{6} = \ln\frac{4}{3} - \frac{1}{6}.$$

□

Question 17. A tent has a circular base of radius $a = 2$ metres and is supported by a horizontal ridge bar held at height $b = 3$ metres above a diameter of the base by vertical supports at each end of the diameter. The material of the tent is stretched tight so that each cross-section perpendicular to the ridge bar is an isosceles triangle. Find the volume of the tent.



(A) $V = \pi$ (B) $V = 3\pi$ (C) None of them (D) $V = 6\pi$

Solution. The correct answer (D). Let the $x-$axis be the diameter of the base under the ridge bar.

The cross-section at position x has base length

$$2\sqrt{a^2 - x^2},$$

so its area is

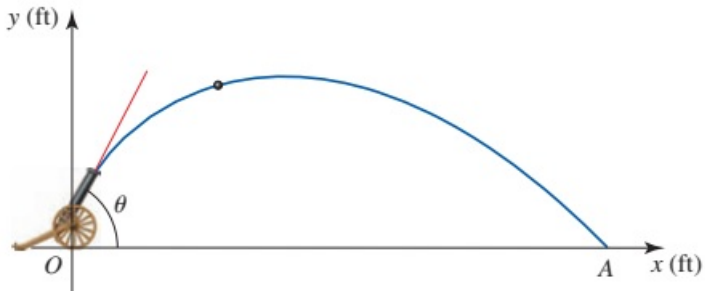
$$A(x) = \frac{1}{2}\left(2\sqrt{a^2 - x^2}\right)b = b\sqrt{a^2 - x^2}.$$

Thus, the volume of the solid is

$$V = \int_{-a}^{a} b\sqrt{a^2 - x^2}dx = b\int_{-a}^{a} \sqrt{a^2 - x^2}dx = b \cdot \frac{\pi a^2}{2} = 3 \times \frac{\pi \times 2^2}{2} = 6\pi$$

□

Question 18. A projectile is fired from a cannon located on a horizontal plane. If we think of the cannon as being located at the origin O of an $xy-$coordinate system, then the path of the projectile is $y = \sqrt{3}.x - \frac{x^2}{400}$; where x and y are measured in feet. Estimate the distance traveled by the projectile in the air, rounded to two decimal numbers.



(A) $L \approx 955.21$ (B) $L \approx 957.21$ (C) None of them (D) $L \approx 956.21$

Solution. The correct answer (D). Points of intersection of the path and $x-$axis (the ground) can be found by solving the equation $\sqrt{3}.x - \frac{x^2}{400} = 0 \Leftrightarrow \left[\begin{array}{c} x = 0 \\ x = 400\sqrt{3} \end{array}\right.$

The required distance is

$$L = \int_0^{400\sqrt{3}} \sqrt{1 + (y')^2}dx = \int_0^{400\sqrt{3}} \sqrt{1 + \left(\sqrt{3} - \frac{x}{200}\right)^2}dx \approx 956.21$$

□

Question 19. Calculate $I = \int_0^{+\infty} \frac{\sqrt{1+x}}{2+7x}\,\mathrm{d}x$.

(A) $I = 0$ (B) $I = +\infty$ (C) $I = \ln 2$ (D) $I = \ln 3$

Solution. The correct answer (B). Let $t = \sqrt{1+x} \Rightarrow t^2 = 1 + x \Rightarrow dx = 2tdt$.

$$I = \int_1^{+\infty} \frac{2t^2}{2 + 7(t^2 - 1)}\,\mathrm{d}t = \int_1^{+\infty} \frac{2}{7} + \frac{10}{49(t^2 - 5/7)}\,\mathrm{d}t = \frac{2}{7}t + \frac{5}{49\sqrt{5/7}} \ln\left|\frac{t - \sqrt{5/7}}{t + \sqrt{5/7}}\right|\Bigg|_1^{+\infty} = +\infty$$

□

Question 20. Calculate $I = \int_0^4 \frac{dx}{(8-x)\sqrt{4-x}}$.

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $+\infty$

Solution. The correct answer (A). Let $t = \sqrt{4-x} \Rightarrow t^2 = 4 - x \Rightarrow 2tdt = -dx$, $\begin{array}{c|cc} x & 0 & 4 \\ \hline t & 2 & 0 \end{array}$. We have

$$\int_0^4 \frac{dx}{(8-x)\sqrt{4-x}} = \int_2^0 \frac{-2tdt}{(8-4+t^2)t} = \int_0^2 \frac{2dt}{4+t^2} =$$

$$= 2 \times \frac{1}{2}\arctan\frac{t}{2}\Big|_0^2 = \arctan 1 - \arctan 0 = \frac{\pi}{4}.$$

□

Part II. Essay (4 points, 30 minutes)

Question 01. The Pacific halibut fishery has been modeled by the differential equation

$$\frac{dy}{dt} = ky\left(1 - \frac{y}{M}\right),$$

where $y(t)$ is the biomass (the total mass of the members of the population) in kilograms at time t (measured in years), the carrying capacity is estimated to be $M = 8 \times 10^7$ kg, and $k = 0.71$ per year.

(a) If $y(0) = 2 \times 10^7$ kg, find the biomass a year later.

(b) How long will it take for the biomass to reach 4×10^7 kg?

Solution. Using separable method, we obtain

$$\frac{dy}{dt} = ky\left(1 - \frac{y}{M}\right) \Rightarrow \frac{dy}{y\left(1 - \frac{y}{M}\right)} = kdt \Rightarrow \int \frac{dy}{y\left(1 - \frac{y}{M}\right)} = \int kdt$$

$$\Rightarrow \int \frac{dy}{y} + \frac{1}{M}\int \frac{dy}{1 - \frac{y}{M}} = k\int dt \Rightarrow \ln|y| - \ln\left|1 - \frac{y}{M}\right| = kt + \ln C$$

$$\Rightarrow \frac{y}{\left|1 - \frac{y}{M}\right|} = Ce^{kt}$$

If $t = 0$, then $\dfrac{y(0)}{\left|1 - \frac{y(0)}{M}\right|} = Ce^{0k} = C \Rightarrow C = \dfrac{8}{3} \times 10^7$.

If $t = 1$, then $\dfrac{y(1)}{\left|1 - \frac{y(1)}{M}\right|} = Ce^{k} \Rightarrow y(1) = \dfrac{Ce^k}{1 + \frac{Ce^k}{M}} \approx 32324112.68$

If $y = 4 \times 10^7$, then $\dfrac{4 \times 10^7}{1 - \frac{4 \times 10^7}{M}} = \dfrac{8}{3} \times 10^7 e^{kt}$

$$\Rightarrow e^{kt} = \frac{3 \times 4 \times 10^7}{8 \times 10^7 \times \left(1 - \frac{4 \times 10^7}{M}\right)} = 3 \Rightarrow t = \frac{\ln(3)}{k} \approx 1.5473(\text{year})$$

Question 02. Solve the differential equation $y'' - 5y' + 6y = 6e^{4x}$

Solution.

(a) **Step 1.** Solve the homogeneous equation

$$y'' - 5y' + 6y = 0$$

The characteristic equation $k^2 - 5k + 6 = 0$ has 2 real different roots

$$k_1 = 2, k_2 = 3$$

(b) **Step 2.** The homogeneous solution is

$$y_h = C_1e^{2x} + C_2e^{3x}$$

(c) **Step 3.** Find a particular solution of nonhomogeneous equation

$$y'' - 5y' + 6y = 6e^{4x}.$$

The particular solution has a form $y_p = x^s.e^{4x}.A$. Since $\alpha = 4$ is not the root of characteristic equation then $s = 0$ and $y_p = A.e^{4x}$.

$$\begin{array}{c|cccc}
6\times & y_p & = & Ae^{4x} \\
-5\times & y_p' & = & 4Ae^{4x} \\
1\times & y_p'' & = & 16Ae^{4x} \\
\hline
y_p'' - 5y_p' + 6y_p & = & 2Ae^{4x} = & 6e^{4x}
\end{array}$$

$$\Rightarrow A = 3$$

(d) **Step 4.** The general solution is

$$y_{gen} = y_h + y_p = C_1e^{2x} + C_2e^{3x} + 3e^{4x}.$$